Customer Purchase Behavior & Segmentation Analytics

Uncovering insights from transactional data to guide strategic business decisions.

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Project Overview

Analyze Behavior

3,900 purchases across categories.

Uncover Insights

Spending patterns, segments, preferences.

Guide Decisions

Optimize strategy, product, and subscriptions.

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Dataset Summary

Key Features

- Customer demographics (Age, Gender, Location, Subscription)
- Purchase details (Item, Category, Amount, Season, Size, Color)
- Shopping behavior (Discount, Promo, Previous Purchases, Frequency, Review, Shipping)

Data Overview

- Rows: 3,900
- Columns: 18
- Missing Data: 37 values in Review Rating

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Exploratory Data Analysis (Python)

01

Data Preparation

Loading, initial exploration (df.info(), .describe()).

02

Data Cleaning

Imputed missing Review Rating with median by category.

03

Column Standardization

Renamed to snake_case for readability.

04

Feature Engineering

Created age_group, purchase_frequency_days.

05

Data Consistency

Dropped redundant promo_code_used.

06

Database Integration

Loaded cleaned data to PostgreSQL for SQL analysis.

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Data Analysis (SQL)

Revenue by Gender

Male: $157,890

Female: $75,191

High-Spending Discount Users

839 customers spent above average with discounts.

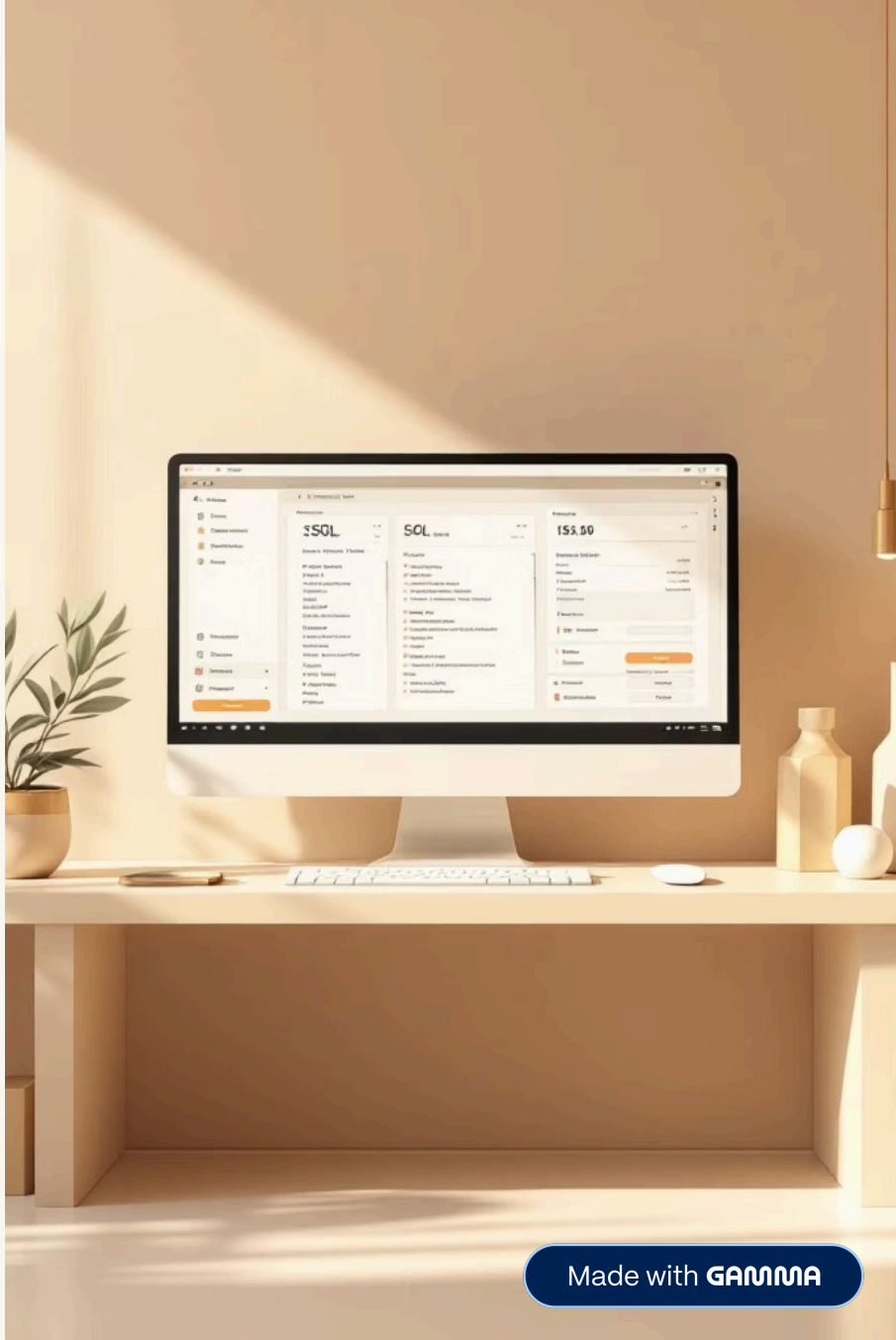
Top 5 Products by Rating

- Gloves (3.86)
- Sandals (3.84)
- Boots (3.82)
- Hat (3.80)
- Skirt (3.78)

Shipping Type Comparison

Express: $60.48 avg. purchase

Standard: $58.46 avg. purchase



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SQL Analysis: Subscriptions & Discounts

Subscribers vs. Non-Subscribers

Subscribers: 1053 customers, $59.49 avg. spend, $62,645 total revenue.

Non-Subscribers: 2847 customers, $59.87 avg. spend, $170,436 total revenue.

Discount-Dependent Products

- Hat (50% discounted)
- Sneakers (49.66% discounted)
- Coat (49.07% discounted)
- Sweater (48.17% discounted)
- Pants (47.37% discounted)

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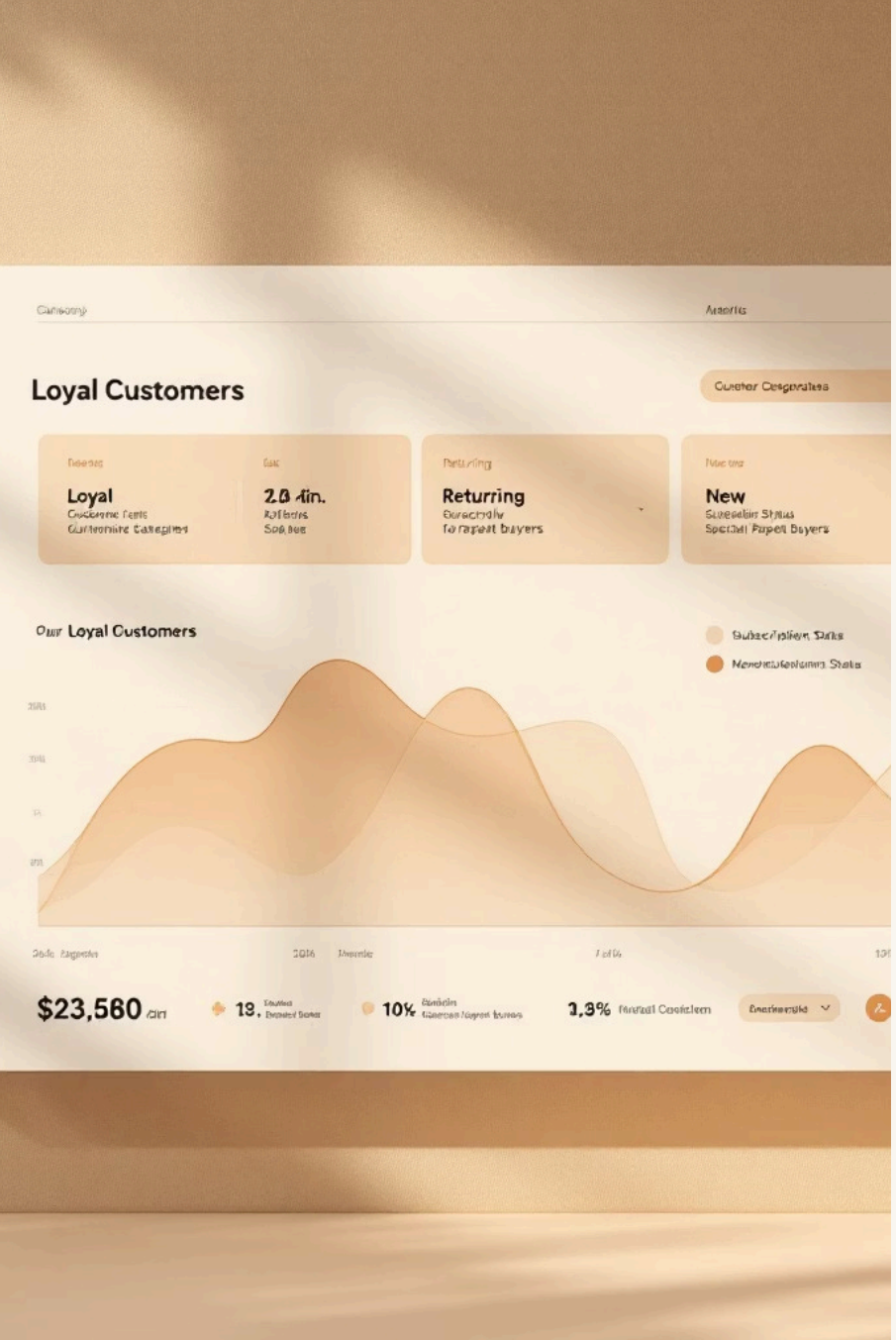

SQL Analysis: Segmentation & Products

Customer Segmentation

- Loyal: 3116 customers
- Returning: 701 customers
- New: 83 customers

Repeat Buyers & Subscriptions

958 repeat buyers are subscribers.

2518 repeat buyers are not subscribers.

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Top Products per Category

Accessories

- Jewelry (171 orders)
- Sunglasses (161 orders)
- Belt (161 orders)

Clothing

- Blouse (171 orders)
- Pants (171 orders)
- Shirt (169 orders)

Footwear

- Sandals (160 orders)
- Shoes (150 orders)
- Sneakers (145 orders)

Outerwear

- Jacket (163 orders)
- Coat (161 orders)

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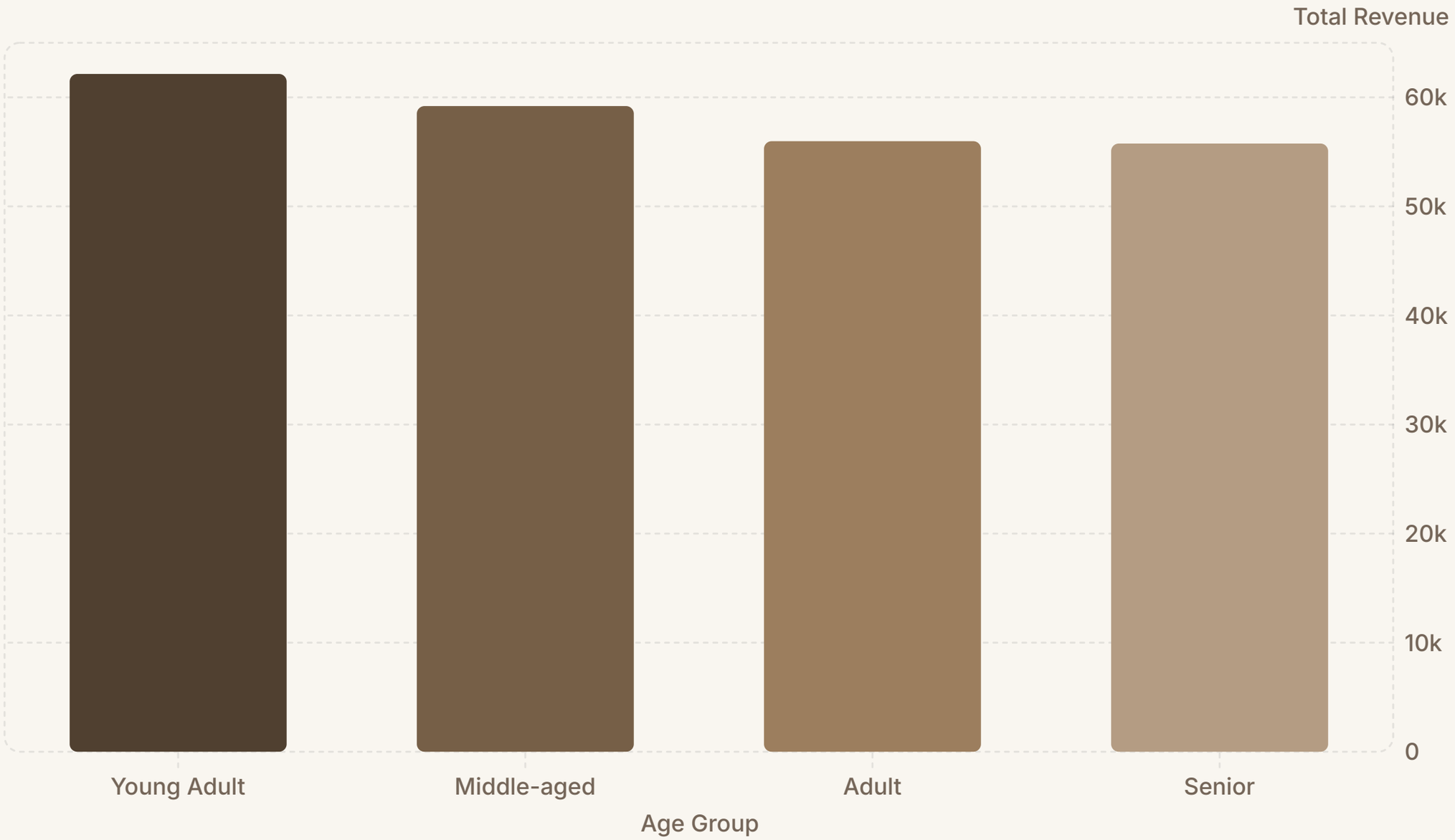
Revenue by Age Group



Young Adults contribute the highest revenue, followed closely by Middle-aged customers.

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Business Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers.

Customer Loyalty

Reward repeat buyers to foster loyalty.

Review Discount Policy

Balance sales boosts with margin control.

Product Positioning

Highlight top-rated and best-selling products.

Targeted Marketing

Focus on high-revenue age groups and express-shipping users.

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